

Mathematics test document

This document is generated for testing how formulas are converted. Each formula below is surrounded by ordinary sentences, as in a real paper, so that the prose can be checked as well.

1 Display formulas

Here is formula d01, of the kind fraction, in its own paragraph.

$$y = \frac{a+b}{c-d}$$

Here is formula d02, of the kind fraction nested, in its own paragraph.

$$z = \frac{1}{1 + \frac{1}{x}}$$

Here is formula d03, of the kind root, in its own paragraph.

$$r = \sqrt{x^2 + y^2}$$

Here is formula d04, of the kind root nth, in its own paragraph.

$$s = \sqrt[3]{ab}$$

The text continues after these formulas with a normal sentence of prose.

Here is formula d05, of the kind scripts, in its own paragraph.

$$E = mc^2$$

Here is formula d06, of the kind scripts scoped, in its own paragraph.

$$P = 10000^{2i/d}$$

Here is formula d07, of the kind scripts both, in its own paragraph.

$$W_i^Q = X_i A$$

Here is formula d08, of the kind greek, in its own paragraph.

$$\alpha + \beta = \gamma\delta$$

The text continues after these formulas with a normal sentence of prose.

Here is formula d09, of the kind sum limits, in its own paragraph.

$$S = \sum_{i=1}^n x_i$$

Here is formula d10, of the kind integral, in its own paragraph.

$$F = \int_0^1 f(x)dx$$

Here is formula d11, of the kind limit, in its own paragraph.

$$L = \lim_{n \rightarrow \infty} a_n$$

Here is formula d12, of the kind fraction root, in its own paragraph.

$$A = \text{softmax}(\frac{QK^T}{\sqrt{d}})V$$

The text continues after these formulas with a normal sentence of prose.
Here is formula d13, of the kind product, in its own paragraph.

$$p = \prod_{k=1}^m q_k$$

Here is formula d14, of the kind matrix, in its own paragraph.

$$M = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

Here is formula d15, of the kind piecewise, in its own paragraph.

$$f(x) = \begin{cases} 1 & x > 0 \\ 0 & x \leq 0 \end{cases}$$

Here is formula d16, of the kind sum fraction, in its own paragraph.

$$\hat{R} = \frac{1}{N} \sum_{j=1}^N \ell(y_j)$$

The text continues after these formulas with a normal sentence of prose.
Here is formula d17, of the kind relations, in its own paragraph.

$$a \leq b \neq c \approx d$$

Here is formula d18, of the kind sets, in its own paragraph.

$$x \in \mathbb{R}^n$$

Here is formula d19, of the kind derivative, in its own paragraph.

$$\frac{\partial L}{\partial w} = -\frac{y}{a}$$

Here is formula d20, of the kind expectation, in its own paragraph.

$$\mu = \mathbb{E}[X] = \int xp(x)dx$$

The text continues after these formulas with a normal sentence of prose.

2 Numbered formulas

The next formula, n01, carries an equation number.

$$\sigma(z) = \frac{1}{1 + e^{-z}} \tag{1}$$

The next formula, n02, carries an equation number.

$$h_t = \tanh(Wh_{t-1} + b) \tag{2}$$

3 A derivation

The following two lines are one aligned derivation.

$$\begin{aligned}(a + b)^2 &= a^2 + 2ab + b^2 \\ &= b^2 + 2ab + a^2\end{aligned}$$

4 Formulas inside sentences

Each term x_i^2 is added to the total.

The scores are scaled by $\frac{1}{\sqrt{d}}$ before the softmax.

We use a weight decay of $\lambda = 0.5$ in all runs.

The weights satisfy $\sum_i w_i = 1$ so they form a distribution.

This holds whenever $n \geq 3$ and fails otherwise.

There are 2^{n-1} such subsets.

5 A table

Quantity	Formula
first	x^2
second	$\sqrt{y}$
third	$\frac{a}{b}$

This is the end of the test document.